

REQUIREMENTS ANALYSIS DOCUMENT for BRAINSTORMING APPLICATION

1.0

Nov 2025

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1 Introduction

1.1 Purpose of the system

The Collaborative Brainstorming Platform is a web-based and mobile application designed to facilitate structured brainstorming sessions using the proven **6-3-5 Method**. This method enables six participants to generate three ideas each over five rounds, resulting in 90+ ideas in just 30 minutes.

The system aims to:

- **Digitize** the traditional paper-based 6-3-5 brainstorming method
- **Enable remote collaboration** for distributed teams
- **Provide AI-powered assistance** for idea generation and session insights
- **Track and document** brainstorming sessions with comprehensive audit trails
- **Facilitate event management** for multiple concurrent brainstorming activities

1.2 Scope of the system

- User authentication and role-based access control (Event Manager, Team Leader, Team Member)
- Event creation and management for brainstorming activities
- Topic definition and assignment to events
- Team formation with up to 6 members per team
- Real-time brainstorming sessions implementing the 6-3-5 methodology
- Synchronized round-based idea submission (3 ideas per member per round, 5 rounds total)
- Timer-based round progression (5 minutes per round)
- ChatGPT integration for AI-powered idea generation and session summaries
- Real-time updates using WebSocket technology
- Session logging and audit trail
- Cross-platform support (Web and Mobile - Flutter)
- REST API for all operations

1.3 Objectives and success criteria of the project

Objectives

- ➔ Digitizing the 6-3-5 Methodology: Transferring the traditional paper-based 6-3-5 brainstorming method to a digital environment, enabling full automation and a structured workflow.
- ➔ Enabling Real-Time Collaboration: Enabling distributed teams from different locations to simultaneously participate in the same brainstorming session via web and mobile platforms.

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- Providing Comprehensive Event Management: Providing a management interface that allows Event Managers to efficiently create, manage, and oversee topics, participants, and teams.
- Implementing Role-Based Access: Clearly segregating system functionality among the Event Manager, Team Leader, and Team Member roles, ensuring each role can only perform operations within its authority.
- Integrating AI Support: Integrating an AI model such as ChatGPT into the system to support idea generation and session summarization processes.
- Providing Automatic Recording and Reporting: Ensuring that all ideas, participant contributions, and session flow (timings, etc.) are automatically recorded and can be reviewed after the session.

Success Criteria

- When a "6-3-5 Session" is initiated, the system must be able to seamlessly manage 6 participants entering 3 ideas per round for 5 rounds (with a 5-minute timer per round), with ideas automatically passed to the next participant between rounds.
- When a team member submits an idea from the web interface, all other web and mobile clients in the same session must be able to see the idea with a delay of less than 2 seconds (via WebSocket/SignalR).
- An "Event Manager" must be able to create a new event and topic, assign 6 participants to this event, and form a team of these participants in less than 5 minutes.
- A user with the "Team Member" role must not be able to access the "Event Manager" panel. A Team Leader should only be able to initiate a session for their own team and should not be able to interfere with another team.
- When a Team Leader clicks the "Generate Idea" button, the system should be able to propose at least three new and meaningful ideas related to the current topic using the ChatGPT API. At the end of the session, a summary should be generated that includes the keywords for all ideas.
- When a session is completed, the system should automatically create an audit trail that includes which idea was submitted by whom in which round and the total duration of the session, and submit it to the Event Manager.

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1.4 Definitions, acronyms, and abbreviations

Term	Definition
6-3-5 Method	A structured brainstorming technique where 6 participants develop 3 ideas in 5 rounds
Event	A planned brainstorming activity with a specific timeframe
Topic	A subject or problem statement for brainstorming
Session	An active brainstorming meeting following the 6-3-5 method
Round	A time-boxed period (5 minutes) within a session for idea generation
Idea	A creative suggestion or solution submitted during a round
Team	A group of up to 6 members participating in brainstorming sessions
AI Annotation	ChatGPT-generated feedback or enhancement to a submitted idea
Event Manager	User role with full system access for event creation and management
Team Leader	User role responsible for managing a specific team

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Team Member	User role for active participants in brainstorming sessions
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1.5 References

- Rohrbach, B. (1969). "Creative by rules – Method 635, a new technique for solving problems." *Absatzwirtschaft* 12: 73-76.
- W3C Web Content Accessibility Guidelines (WCAG) 2.1
- OpenAPI Specification 3.0
- IEEE Std 830-1998 - IEEE Recommended Practice for Software Requirements Specifications
- UML 2.5 Specification - Object Management Group (OMG)

1.6 Overview

This Requirements Analysis Document is organized into five main sections:

- **Section 2** describes the current state of brainstorming practices and their limitations.
- **Section 3** presents the proposed system with detailed functional and non-functional requirements, accompanied by UML models
- **Section 4** provides a comprehensive glossary of domain-specific terms
- **Section 5** establishes traceability between requirements and use cases
- **Section 6** includes appendices with additional technical details

2 Current system

2.1 Current State of Affairs

Currently, organizations and teams conduct brainstorming sessions using traditional methods:

Physical Meetings:

- Teams gather in meeting rooms with whiteboards, sticky notes, or flip charts
- A facilitator manages the session timing and idea collection
- Ideas are manually recorded on paper or photographed after sessions
- Physical presence is required, limiting participation from remote team members

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Digital Collaboration Tools:

- Generic tools like Miro, Mural, or shared documents are used
- No built-in support for structured methods like 6-3-5
- Manual tracking of rounds and timing
- Lack of enforcement for the 6-3-5 rules (3 ideas per round, 6 participants)
- Limited AI assistance for idea generation

Ad-hoc Solutions:

- Some organizations use spreadsheets or custom scripts
- No centralized system for managing multiple events or teams
- Difficulty in tracking historical sessions and outcomes
- No audit trail or session analytics

2.2 Problems with Current System

1. **Lack of Structure:** Generic tools don't enforce the 6-3-5 methodology, leading to inconsistent execution
2. **Manual Overhead:** Facilitators must manually track time, count ideas, and manage round transitions
3. **Limited Remote Support:** Physical methods exclude remote participants; digital tools lack real-time synchronization
4. **Scalability Issues:** Managing multiple concurrent brainstorming events is challenging
5. **Missing AI Assistance:** No intelligent support for idea generation or session insights
6. **Accessibility:** Physical methods exclude team members with mobility constraints
7. **Data Loss Risk:** Paper-based sessions can be lost; photos of whiteboards are difficult to search

2.3 Limitations

- **Participation Barriers:** Introverted members may not contribute equally in face-to-face settings
- **No Real-time Validation:** Facilitators can't enforce the "3 ideas per round" rule effectively
- **Resource Intensive:** Requires dedicated meeting rooms and facilitation time

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3 Proposed system

3.1 Overview

The Collaborative Brainstorming Platform addresses the limitations of current systems by providing a comprehensive, structured, and AI-enhanced solution for conducting brainstorming sessions using the 6-3-5 method.

Key Features:

- **Structured Workflow:** Enforces 6-3-5 rules automatically (6 members, 3 ideas, 5 rounds)
- **Real-time Collaboration:** WebSocket-based synchronization for all participants
- **AI Integration:** ChatGPT-powered idea generation and session summaries
- **Multi-platform:** Web and Mobile applications
- **Role-based Access:** Three distinct roles with appropriate permissions
- **Comprehensive Audit:** Complete logging of all actions and session history

3.2 Functional requirements

User Management

FR-1: The system shall allow users to register with email, password, first name, last name, phone number (optional), and role (Event Manager, Team Leader, or Team Member).

FR-2: The system shall validate that email addresses are unique and properly formatted during registration.

FR-3: The system shall store user passwords securely using an irreversible encryption mechanism.

FR-4: The system shall allow registered users to log in using their email and password.

FR-5: The system shall generate a secure session token upon successful login to authenticate subsequent requests.

FR-6: The system shall require a valid session token for all operations that access protected resources.

FR-7: The system shall allow users to view their own profile information.

FR-8: The system shall allow users to update their profile information (first name, last name, phone number).

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FR-9: The system shall prevent users from changing their email address after registration.

FR-10: The system shall enforce role-based access control for all system operations.

Event Management

FR-11: The system shall allow Event Managers to create new events with name, description, start date, and end date.

FR-12: The system shall validate that the end date is after the start date for all events.

FR-13: The system shall automatically set the event status to "Planned" upon creation.

FR-14: The system shall allow Event Managers to update event details (name, description, dates).

FR-15: The system shall allow Event Managers to change event status (Planned, Active, Completed, Cancelled).

FR-16: The system shall allow Event Managers to delete events that have no associated sessions.

FR-17: The system shall allow all authenticated users to view a list of all events.

FR-18: The system shall allow users to filter events by status.

FR-19: The system shall display event details including name, description, dates, status, and creator information.

FR-20: The system shall prevent deletion of events that have associated brainstorming sessions.

Topic Management

FR-21: The system shall allow Event Managers to create topics with title and description.

FR-22: The system shall allow Event Managers to assign topics to specific events.

FR-23: The system shall allow Event Managers to update topic information.

FR-24: The system shall allow Event Managers to change topic status (Open, Closed, Archived).

FR-25: The system shall allow all authenticated users to view topics associated with events they have access to.

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FR-26: The system shall allow Event Managers to delete topics that are not assigned to any sessions.

FR-27: The system shall display all topics associated with a specific event.

Team Management

FR-28: The system shall allow Team Leaders to create teams with name, description, and maximum members (default 6).

FR-29: The system shall enforce a maximum of 6 members per team for 6-3-5 methodology compliance.

FR-30: The system shall allow Team Leaders to assign a team leader to each team.

FR-31: The system shall allow Team Leaders to add members to their teams.

FR-32: The system shall validate that a member can only be added if the team has not reached maximum capacity.

FR-33: The system shall allow Team Leaders to remove members from their teams.

FR-34: The system shall prevent removal of team members if they have active sessions.

FR-35: The system shall allow Team Leaders to view all members of their teams.

FR-36: The system shall display team information including name, description, leader, and member count.

FR-37: The system shall allow Event Managers to assign teams to events.

Event Participation

FR-38: The system shall allow Event Managers to add participants to events.

FR-39: The system shall allow Event Managers to remove participants from events.

FR-40: The system shall send notifications to users when they are added to an event.

FR-41: The system shall allow users to view all events they are participating in.

FR-42: The system shall display participant lists for each event.

FR-43: The system shall prevent duplicate participation records for the same user and event.

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Brainstorming Session Management

FR-44: The system shall allow Team Leaders to create brainstorming sessions by selecting a team and topic.

FR-45: The system shall validate that the selected team has at least 3 members before creating a session.

FR-46: The system shall initialize sessions with status "NotStarted" and round number 0.

FR-47: The system shall configure all Brainstorming Sessions to adhere to the core 6-3-5 methodology structure.

FR-48: The system shall allow Team Leaders to start a session, changing status to "InProgress".

FR-49: The system shall automatically create the first round when a session is started.

FR-50: The system shall start a countdown timer when a round begins.

FR-51: The system shall broadcast the timer countdown to all session participants in real-time.

FR-52: The system shall allow Team Leaders to pause an active session.

FR-53: The system shall allow Team Leaders to resume a paused session.

FR-54: The system shall automatically advance to the next round when the timer expires.

FR-55: The system shall validate that all team members have submitted exactly 3 ideas before allowing manual round advancement.

FR-56: The system shall mark a session as "Completed" after 5 rounds are finished.

FR-57: The system shall allow Team Leaders to manually end a session at any time.

FR-58: The system shall prevent starting new rounds if the session has reached 5 rounds.

FR-59: The system shall display session information including team, topic, status, current round, and elapsed time.

FR-60: The system shall allow participants to view the session they are currently in.

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Round Management

FR-61: The system shall automatically create a new round when the previous round completes.

FR-62: The system shall assign a sequential round number (1 to 5) to each round.

FR-63: The system shall record the start time when a round begins.

FR-64: The system shall record the end time when a round completes.

FR-65: The system shall display the current round number to all participants.

FR-66: The system shall show elapsed time and remaining time for the current round.

FR-67: The system shall prevent idea submission after a round has ended.

Idea Management

FR-68: The system shall allow team members to submit ideas during an active round.

FR-69: The system shall validate that each user submits exactly 3 ideas per round.

FR-70: The system shall prevent users from submitting more than 3 ideas in a single round.

FR-71: The system shall prevent users from submitting ideas if they are not part of the session's team.

FR-72: The system shall assign a sequential order (1, 2, 3) to each idea within a round.

FR-73: The system shall record the submission timestamp for each idea.

FR-74: The system shall mark ideas as "AI-generated: false" by default for user-submitted ideas.

FR-75: The system shall allow users to edit their ideas before the round ends.

FR-76: The system shall prevent editing of ideas after the round has ended.

FR-77: The system shall broadcast newly submitted ideas to all session participants in real-time.

FR-78: The system shall display all ideas submitted in the current round to all participants.

FR-79: The system shall allow users to view ideas from previous rounds in the same session.

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FR-80: The system shall display the author's name for each idea.

FR-81: The system shall show the round number and order for each idea.

ChatGPT Integration

FR-82: The system shall provide an option to generate AI-powered ideas during a round.

FR-83: The system shall call the OpenAI API with the topic description to generate ideas.

FR-84: The system shall allow users to request AI-generated ideas when they have not reached the 3-idea limit.

FR-85: The system shall mark AI-generated ideas with "AI-generated: true".

FR-86: The system shall allow users to edit AI-generated ideas before submitting them.

FR-87: The system shall log all ChatGPT API calls including prompt, response, and token usage.

FR-88: The system shall provide AI-powered session summaries after a session completes.

FR-89: The system shall generate AI annotations for ideas suggesting improvements or related concepts.

FR-90: The system shall display AI annotations alongside the original ideas.

FR-91: The system shall rate-limit ChatGPT API calls to prevent abuse (max 10 calls per user per session).

Real-time Communication (SignalR)

FR-92: The system shall establish WebSocket connections for all active session participants.

FR-93: The system shall broadcast "RoundStarted" events when a new round begins.

FR-94: The system shall broadcast "RoundEnded" events when a round completes.

FR-95: The system shall broadcast "IdeaSubmitted" events when a participant submits an idea.

FR-96: The system shall broadcast "SessionStatusChanged" events when session status updates.

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FR-97: The system shall broadcast "ParticipantJoined" events when a user joins a session.

FR-98: The system shall broadcast "ParticipantLeft" events when a user leaves a session.

FR-99: The system shall send "TimerUpdate" events every second during active rounds.

FR-100: The system shall send "Notification" events for important system messages.

FR-101: The system shall automatically reconnect WebSocket connections if they drop.

FR-102: The system shall maintain session state for reconnecting clients.

Session Logging and Audit Trail

FR-103: The system shall log all session-related actions (start, pause, resume, end).

FR-104: The system shall log all round transitions with timestamps.

FR-105: The system shall log all idea submissions with user and timestamp information.

FR-106: The system shall log all ChatGPT interactions with prompts and responses.

FR-107: The system shall record the user who performed each logged action.

FR-108: The system shall allow Event Managers and Team Leaders to view session logs.

FR-109: The system shall provide filtering options for logs (by session, by user, by date range).

FR-110: The system shall export session logs in JSON format.

FR-111: The system shall retain session logs for a minimum of 1 year.

Reporting and Analytics

FR-112: The system shall display the total number of ideas generated per session.

FR-113: The system shall calculate and display the average ideas per participant.

FR-114: The system shall show the distribution of ideas across rounds.

FR-115: The system shall identify the most active participants (by idea count).

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FR-116: The system shall display session duration and time spent per round.

FR-117: The system shall show the percentage of AI-generated ideas vs. user-generated ideas.

FR-118: The system shall allow exporting session results to PDF or Excel format.

3.3 Nonfunctional requirements

Nonfunctional requirements describes user-level requirements that are not directly related to functionality. This includes usability, reliability, performance, supportability, implementation, interface, operational, packaging, and legal requirements.

■ **Usability**

- NFR1: The web interface shall be responsive and work on desktop, tablet, and mobile browsers.
- NFR2: The system shall provide visual feedback for all user actions within 100ms.
- NFR3: The system shall display error messages in a user-friendly format.
- NFR4: The timer display shall be prominently visible during active rounds.
- NFR5: The system shall provide keyboard shortcuts for common actions (submit idea: Ctrl+Enter).

■ **Reliability**

- NFR6: The system shall have an uptime of 99.5% (excluding planned maintenance).
- NFR7: The system shall gracefully handle WebSocket connection failures with automatic reconnection.
- NFR8: The system shall persist session state to the database every 30 seconds to prevent data loss.
- NFR9: The system shall implement database backups every 24 hours with 30-day retention.
- NFR10: The system shall log all errors with stack traces for debugging purposes.

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■ **Performance**

- NFR11: The system shall support at least 100 concurrent brainstorming sessions.
- NFR12: The system shall handle at least 600 concurrent WebSocket connections (100 sessions × 6 users).
- NFR13: The system shall respond to REST API requests within 500ms under normal load.
- NFR14: The system shall broadcast real-time events to all session participants within 100ms.
- NFR15: The system shall process and store an idea submission within 200ms.
- NFR16: The ChatGPT API integration shall timeout after 10 seconds if no response is received.

■ **Supportability**

- NFR17: Each API request must carry a unique "Correlation ID" to allow tracking across all system layers (frontend, backend, database, AI service).
- NFR18: The backend API must provide a /health API endpoint that monitors the overall health of the system (database connection, AI API access).

■ **Implementation**

- The project's implementation will be built on a modern software architecture that supports cross-platform compatibility and easy scalability. The solution will consist of a well-defined, standards-compliant RESTful API that manages all operations, and modern front-end applications that communicate with this API, built on a single, shared codebase. To ensure ease of deployment and consistency, all back-end components will be packaged using containerization technology (Docker) and automated CI/CD processes will be adopted for rapid release cycles.

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■ **Interface**

- NFR-19: The system must integrate with the OpenAI GPT-4 (or later) REST API for all AI functions (idea generation, summarization).
- NFR-20: The system's own REST API (backend) must comply with the Swagger specification to ensure consistent communication with web and mobile clients.
- NFR-21: The system must allow users to register and log in using their Google or Microsoft accounts (using the OAuth 2.0 protocol).

■ **Packaging**

- NFR-22: Backend API and web frontend applications should be packaged as Docker containers to standardize deployment.
- NFR-23: The mobile application should be packaged as signed .ipa and .apk (or .aab) files that can be uploaded to the Apple App Store and Google Play Store.
- NFR-24: All packaging processes should be automated through a CI/CD (Continuous Integration/Continuous Delivery) pipeline (e.g., GitHub Actions).

■ **Legal**

- NFR-25 (User Consent): The system must provide a checkbox on the registration screen stating "I have read and agree to the Terms of Use and Privacy Policy."
- NFR-26 (Registration Block): The system must prevent registration from being completed without checking this checkbox.
- NFR-27 (Licensing): The project should aim to use only third-party libraries with open source and permissive licenses, such as MIT, Apache 2.0, or BSD.

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3.4 System models

■ **Scenarios**

3.4.1.1 Create Brainstorming Session

Primary Actor: Team Leader

Scope: Brainstorming Application

Level: User Goal

Stakeholders:

- Team Leader: Wants to initiate a structured brainstorming session
- Team Members: Want to participate in the session
- Event Manager: Wants to track event-related sessions

Preconditions:

- Team Leader is authenticated
- At least one team exists with 3-6 members
- At least one topic exists and is assigned to an event

Success Guarantee (Postconditions):

- A new brainstorming session is created with status "NotStarted"
- Session is linked to the selected team and topic
- Session is configured for 5 rounds with 5-minute duration per round
- All team members are notified of the new session

Main Success Scenario:

1. Team Leader navigates to "Create Session" page
2. System displays a form with available teams and topics
3. Team Leader selects a team from the dropdown list
4. Team Leader selects a topic from the dropdown list
5. Team Leader clicks "Create Session" button
6. System validates that the selected team has between 3 and 6 members
7. System validates that the selected topic is active and available
8. System creates a new session record with:
 - i. Team ID
 - ii. Topic ID
 - iii. Status: "NotStarted"
 - iv. Current Round: 0
 - v. Total Rounds: 5
 - vi. Round Duration: 5 minutes
 - vii. Created timestamp
9. System saves the session to the database
10. System displays success message with session details
11. System navigates Team Leader to the session lobby page
12. System sends real-time notification to all team members

3.4.1.2 Start Session

Primary Actor: Team Leader

Scope: Brainstorming Application

Level: User Goal

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Stakeholders:

- Team Leader: Wants to begin the brainstorming process
- Team Members: Want to start contributing ideas
- System: Needs to initialize round tracking and timers

Preconditions:

- Team Leader is authenticated
- A session exists with status "NotStarted" or "Paused"
- All team members are present in the session lobby

Success Guarantee (Postconditions):

- Session status changes to "InProgress"
- First round (Round 1) is created and activated
- 5-minute countdown timer starts
- All participants see the active session interface
- Real-time updates are enabled for all participants

Main Success Scenario:

1. Team Leader is on the session lobby page
2. System displays session details and "Start Session" button
3. System shows list of team members who have joined
4. Team Leader clicks "Start Session" button
5. System validates that at least 3 team members are present
6. System updates session status to "InProgress"
7. System records session start timestamp
8. System creates Round 1 with:
 - i. Round Number: 1
 - ii. Session ID
 - iii. Start Time: current timestamp
 - iv. Status: "Active"
9. System starts a 5-minute countdown timer (300 seconds)
10. System broadcasts "SessionStarted" event to participants
11. System displays active session UI with:
 - i. Topic description
 - ii. Current round number (1/5)
 - iii. Timer countdown
 - iv. Idea submission form
 - v. Participant list with online status
12. System enables idea submission for all participants

3.4.3 Submit Idea

Primary Actor: Team Member

Scope: Brainstorming Application

Level: User Goal

Stakeholders:

- Team Member: Wants to contribute an idea
- Other Team Members: Want to see ideas as they are submitted
- Team Leader: Wants to track participation

Preconditions:

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- Team Member is authenticated
- Team Member is part of an active session
- Current round is in progress (timer running)
- Team Member has submitted fewer than 3 ideas in current round

Success Guarantee (Postconditions):

- Idea is saved to the database
- Idea is immediately visible to all session participants
- Idea counter for the user is incremented
- If this is the 3rd idea, submission form is disabled for this user

Main Success Scenario:

1. Team Member is on the active session page
2. System displays idea submission form with character counter (max 500 characters)
3. System shows "Ideas submitted: X/3" counter
4. Team Member types an idea into the text area
5. Team Member clicks "Submit Idea" button
6. System validates idea content (not empty, ≤ 500 characters)
7. System validates that user has not exceeded 3 ideas for current round
8. System validates that the round is still active
9. System creates new Idea record with:
 - a. Round ID
 - b. Session ID
 - c. User ID
 - d. Content (the idea text)
 - e. Order in Round (1, 2, or 3)
 - f. Submission timestamp
 - g. isAIGenerated: false
10. System saves idea to database
11. System updates user's idea count: $X+1/3$
12. System broadcasts "IdeaSubmitted" event with idea details
13. System displays the new idea in the ideas list for all participants
14. System clears the idea input form
15. If user has submitted 3 ideas:
 - a. System disables submission form
 - b. System displays "You've submitted all 3 ideas for this round"
16. System displays success notification

3.4.4 Request AI Idea Generation

Primary Actor: Team Member

Scope: Brainstorming Application

Level: User Goal

Stakeholders:

- Team Member: Wants AI assistance for idea generation
- Organization: Wants to track AI usage for billing
- OpenAI: Provides the AI service

Preconditions:

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- Team Member is authenticated
- Team Member is in an active session
- Team Member has submitted fewer than 3 ideas in current round
- User has not exceeded AI request limit (10 per session)

Success Guarantee (Postconditions):

- AI-generated idea is displayed to the user
- User can edit and submit the AI idea
- ChatGPT interaction is logged for audit
- Token usage is recorded

Main Success Scenario:

1. Team Member is on the active session page with idea submission form
2. System displays "Generate AI Idea" button
3. System shows AI usage counter: "X/10 AI requests used"
4. Team Member clicks "Generate AI Idea" button
5. System disables the button and shows loading spinner
6. System retrieves the session's topic description
7. System calls OpenAI GPT-4 API with the prompt
8. System waits for API response (timeout: 10 seconds)
9. System receives AI-generated idea text
10. System logs the ChatGPT interaction with:
 - i. Session ID
 - ii. User ID
 - iii. Prompt sent
 - iv. Response received
 - v. Token usage (prompt + completion)
 - vi. Timestamp
11. System increments user's AI request counter
12. System populates the idea submission form with AI-generated text
13. System marks the text area with visual indicator "AI-Generated - Edit before submitting"
14. System enables "Submit Idea" button
15. Team Member reviews and edits the AI idea (optional)
16. Team Member follows UC-28 (Submit Idea) to submit

3.4.5 Advance to Next Round

Primary Actor: System Timer (automatic) or Team Leader (manual)

Scope: Brainstorming Application

Level: System Function

Stakeholders:

- Team Members: Want smooth transition to next round
- Team Leader: Wants to track progress through rounds
- System: Needs to maintain session integrity

Preconditions:

- Session is in "InProgress" status
- Current round is active
- Either: 5-minute timer has expired OR Team Leader manually advances

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Success Guarantee (Postconditions):

- Current round is marked as completed
- New round is created (if not at round 5)
- Timer resets to 5 minutes for new round
- Idea submission counters reset to 0/3 for all users
- All participants are notified of round transition

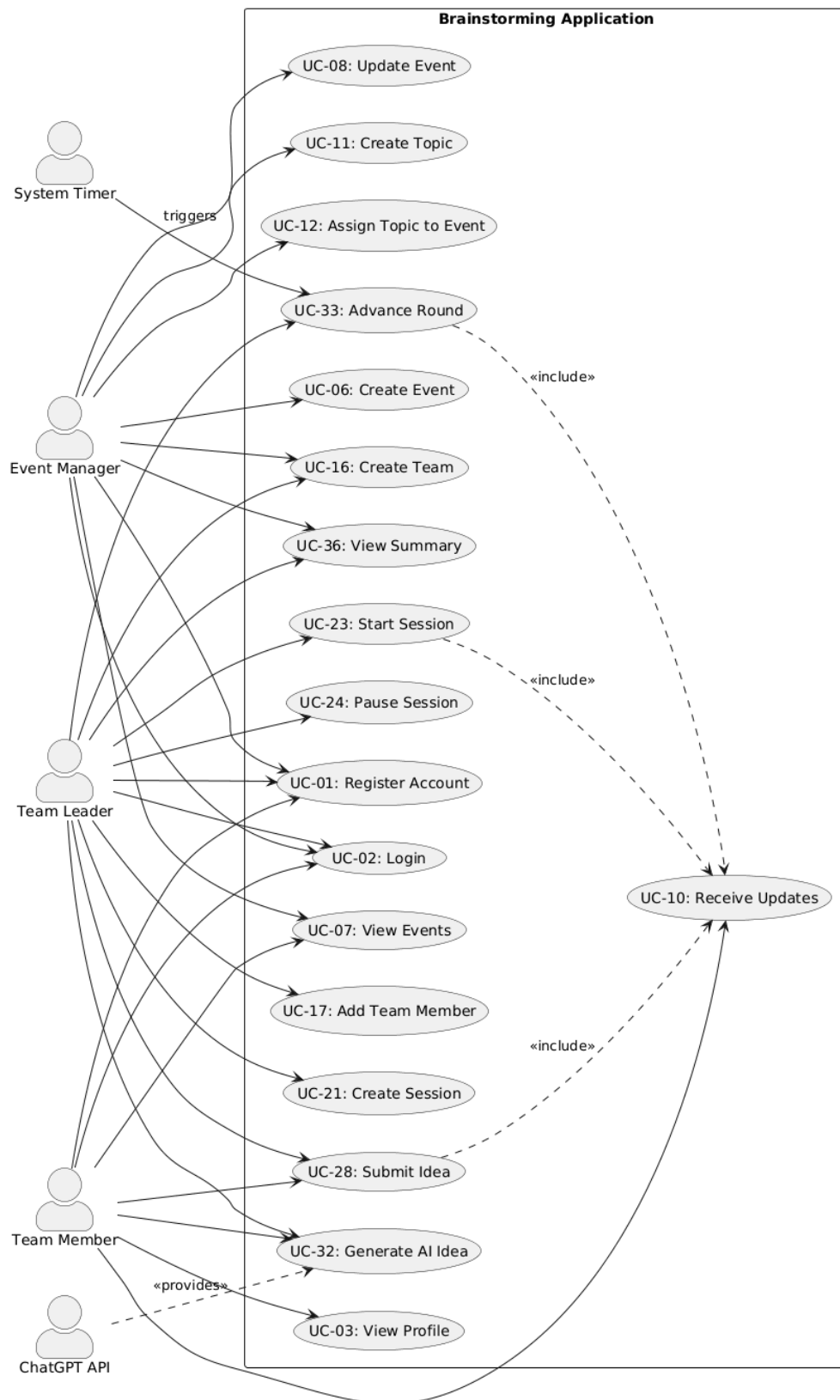
Main Success Scenario (Automatic Advance):

1. System timer reaches 00:00 (5 minutes elapsed)
2. System validates that current round is still active
3. System updates current round with:
 - a. End Time: current timestamp
 - b. Status: "Completed"
4. System checks current round number
5. If current round < 5:
6. System creates new round with:
 - a. Round Number: current + 1
 - b. Session ID
 - c. Start Time: current timestamp
 - d. Status: "Active"
7. System updates session's currentRound field
8. System resets countdown timer to 300 seconds
9. System broadcasts "RoundEnded" event
10. System broadcasts "RoundStarted" event with new round details
11. System displays round transition animation to all participants
12. System updates UI to show new round number (e.g., "2/5")
13. System resets idea submission counters to 0/3 for all users
14. System re-enables idea submission forms
 - a. If current round = 5
15. System updates session status to "Completed"
16. System records session end timestamp
17. System broadcasts "SessionCompleted" event
18. System navigates all participants to session summary page
19. System generates session statistics

Alternative Flow (Manual Advance by Team Leader):

1. Team Leader clicks "Advance Round" button
2. System displays confirmation dialog: "Some members haven't submitted all ideas. Advance anyway?"
3. Team Leader confirms
4. Use case continues from Main Scenario step 2

■ Use case model

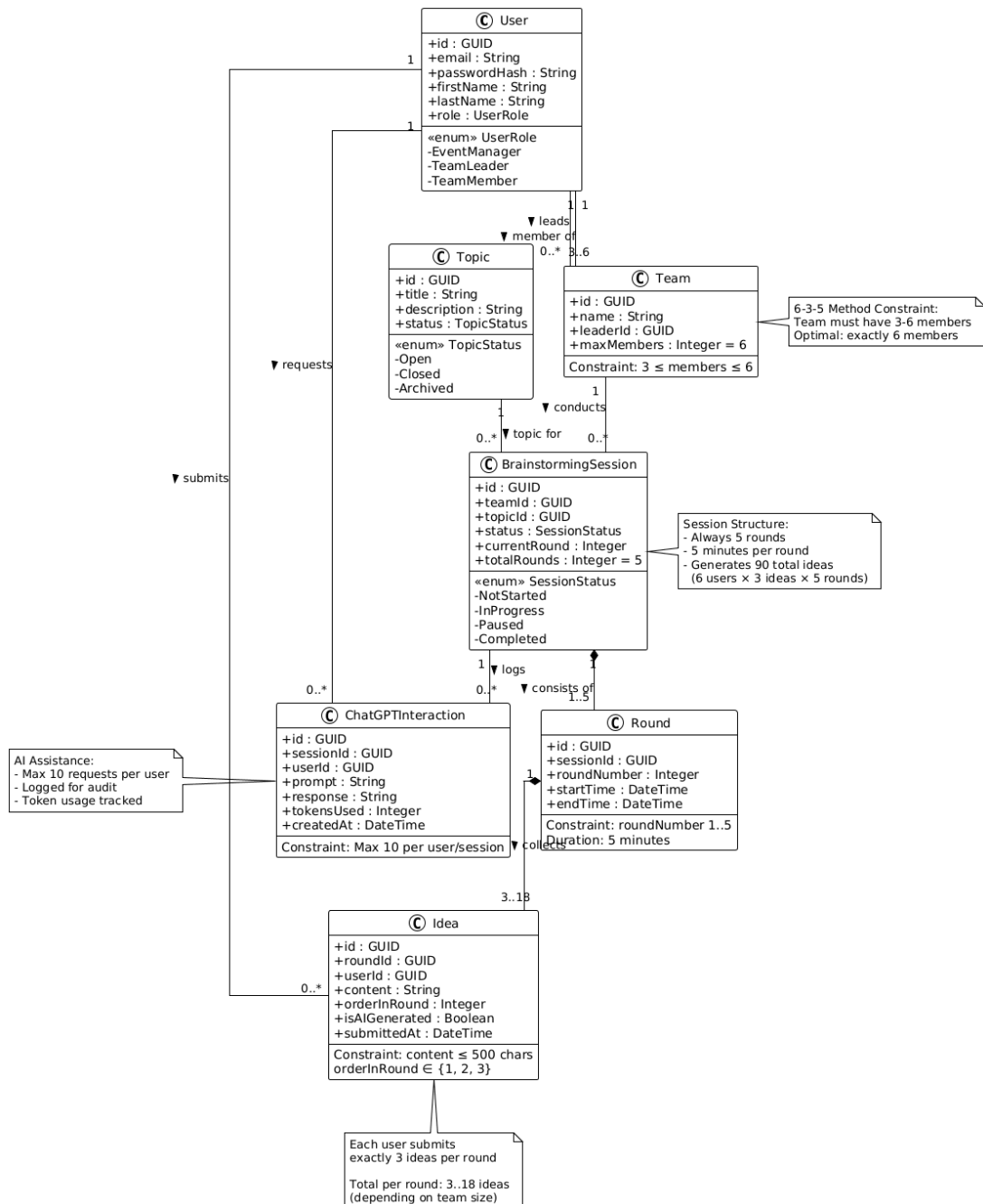


■ Object model

3.4.1 Overview

This domain model represents the conceptual structure of the Brainstorming Application at the analysis level. It shows entities, their attributes, and relationships without implementation details (no methods, no database-specific elements).

3.4.2 Class Diagram



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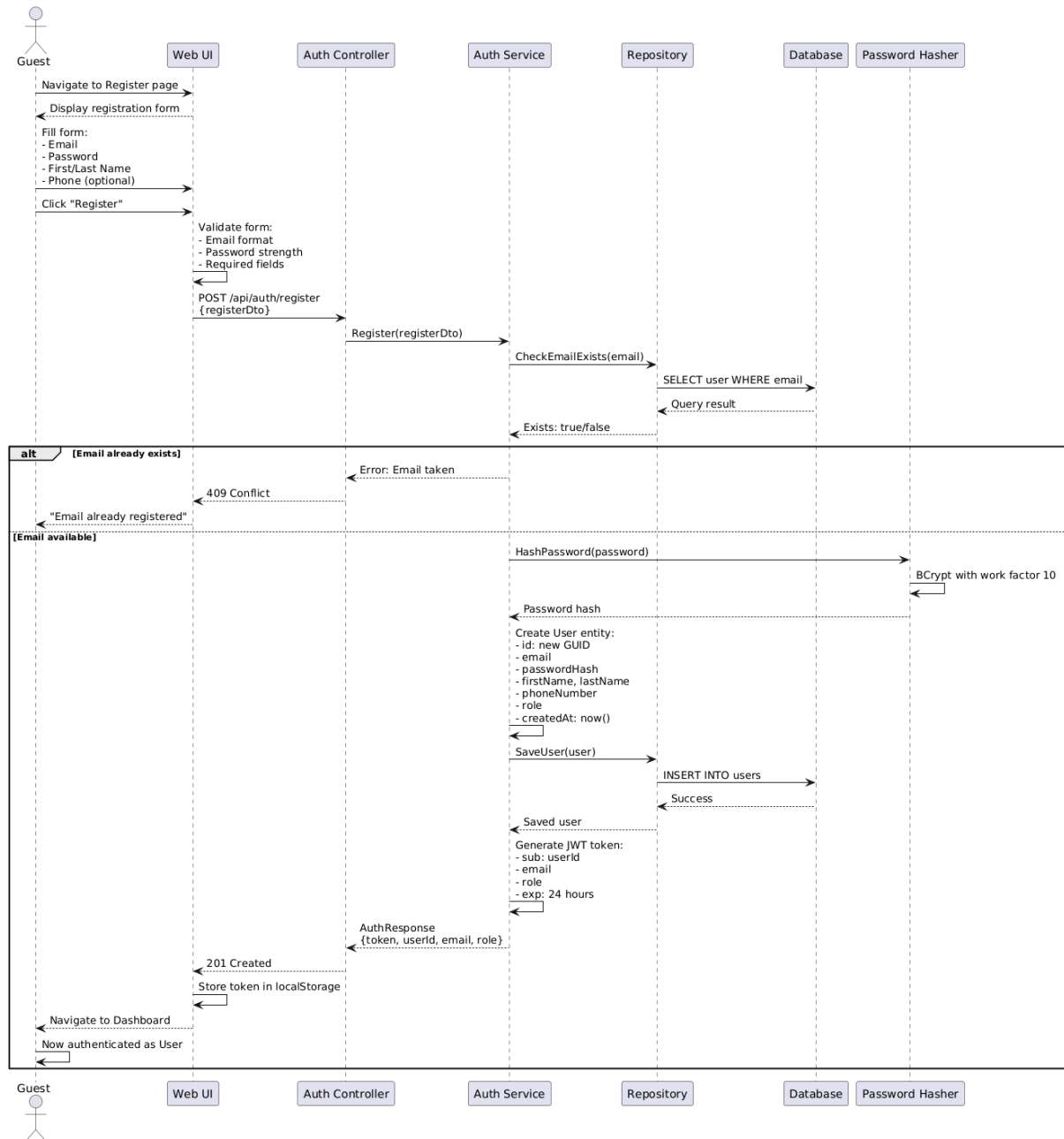
3.4.■.3 *Entity Descriptions*

- 3.4.■.3.1 User:** Represents a registered participant of the system. Each user can have one of the defined roles determining their permissions and responsibilities in the system. Users may create, organize, or participate in events, as well as lead teams and submit ideas during brainstorming sessions.
- 3.4.■.3.2 Topic:** Represents a subject, problem statement, or question discussed during brainstorming sessions within an event. Each topic belongs to one event and is led by a user. It defines the thematic focus for teams to generate ideas.
- 3.4.■.3.3 Team:** Represents a group of users collaborating in brainstorming sessions. Each team operates under a leader and conducts sessions on specific topics.
- 3.4.■.3.4 BrainstormingSession:** Represents an active or completed brainstorming process following the 6-3-5 method. Each session is linked to a topic and a team, and it consists of multiple rounds where members generate and exchange ideas.
- 3.4.■.3.5 Round:** Represents a 5-minute time-boxed period within a brainstorming session where users generate new ideas. Each session consists of exactly five rounds.
- 3.4.■.3.6 Idea:** Represents a creative suggestion generated by a user during a brainstorming round. Each user submits three ideas per round, which may be human- or AI-generated.
- 3.4.■.3.7 ChatGPTInteraction:** Logs all AI-related interactions during brainstorming sessions. Each record stores user prompts, model responses, and token usage for audit and monitoring purposes.

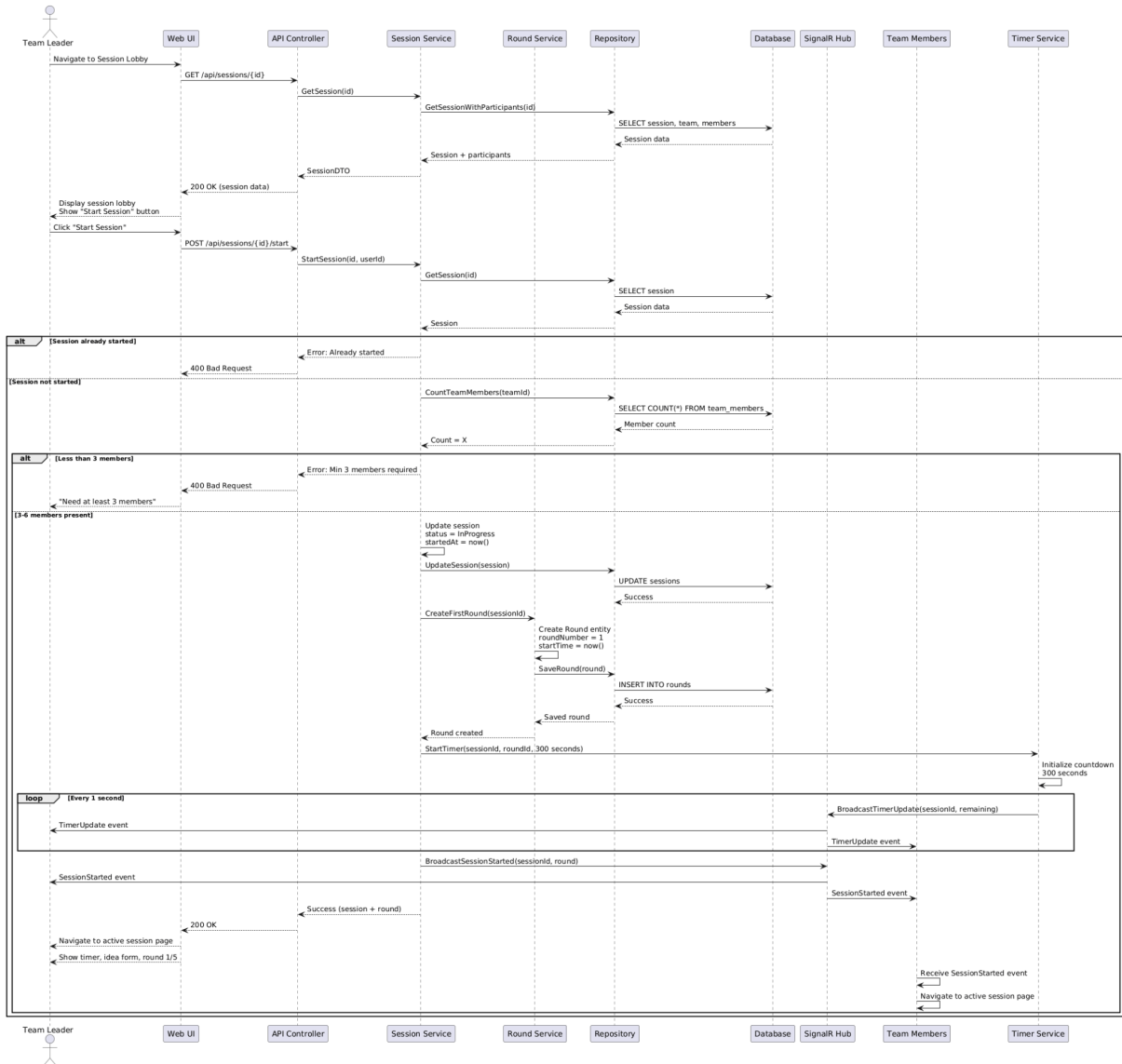
■ *Dynamic model*

3.4.1.1 Sequence Diagrams

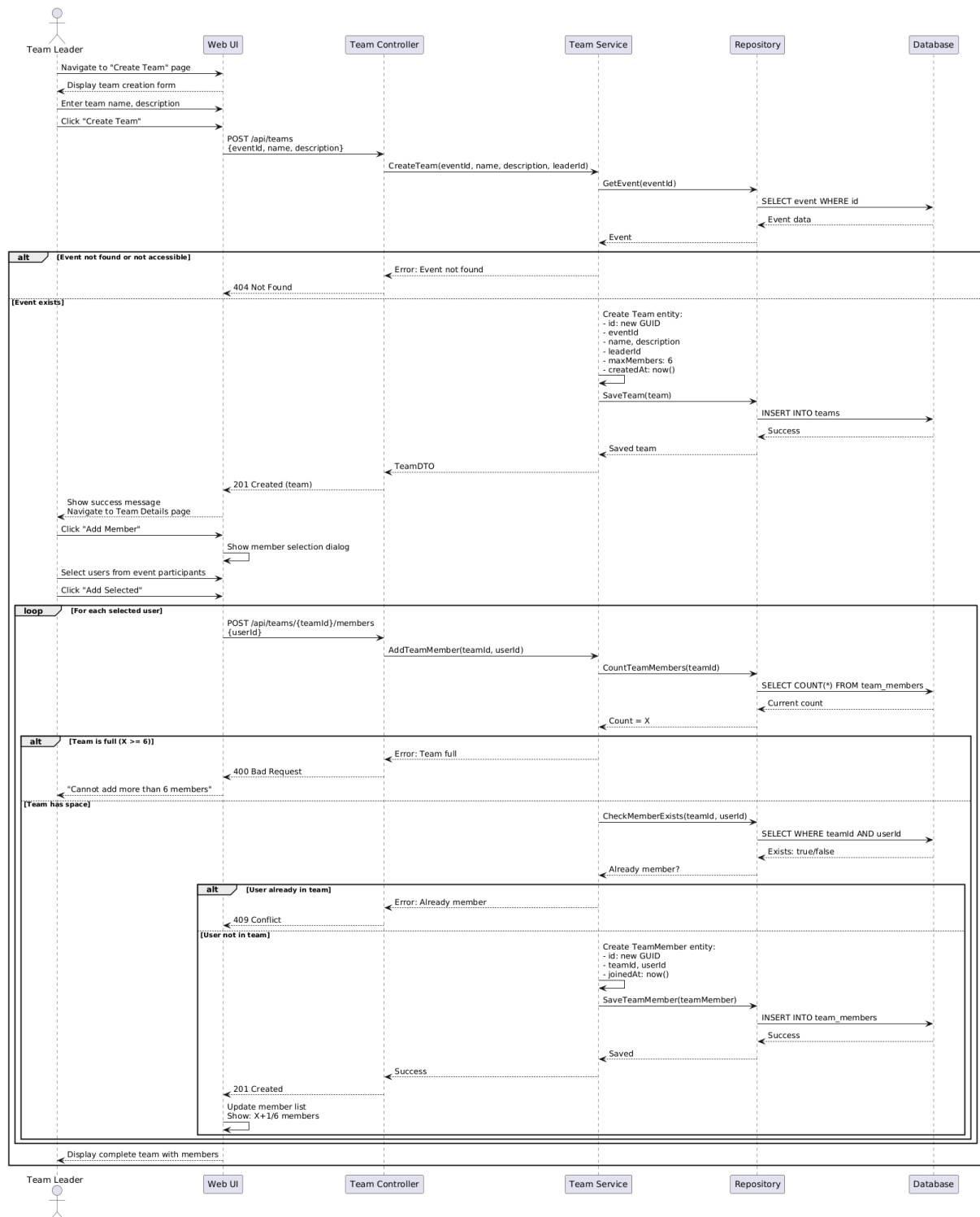
3.4.1.1.1 User Registration



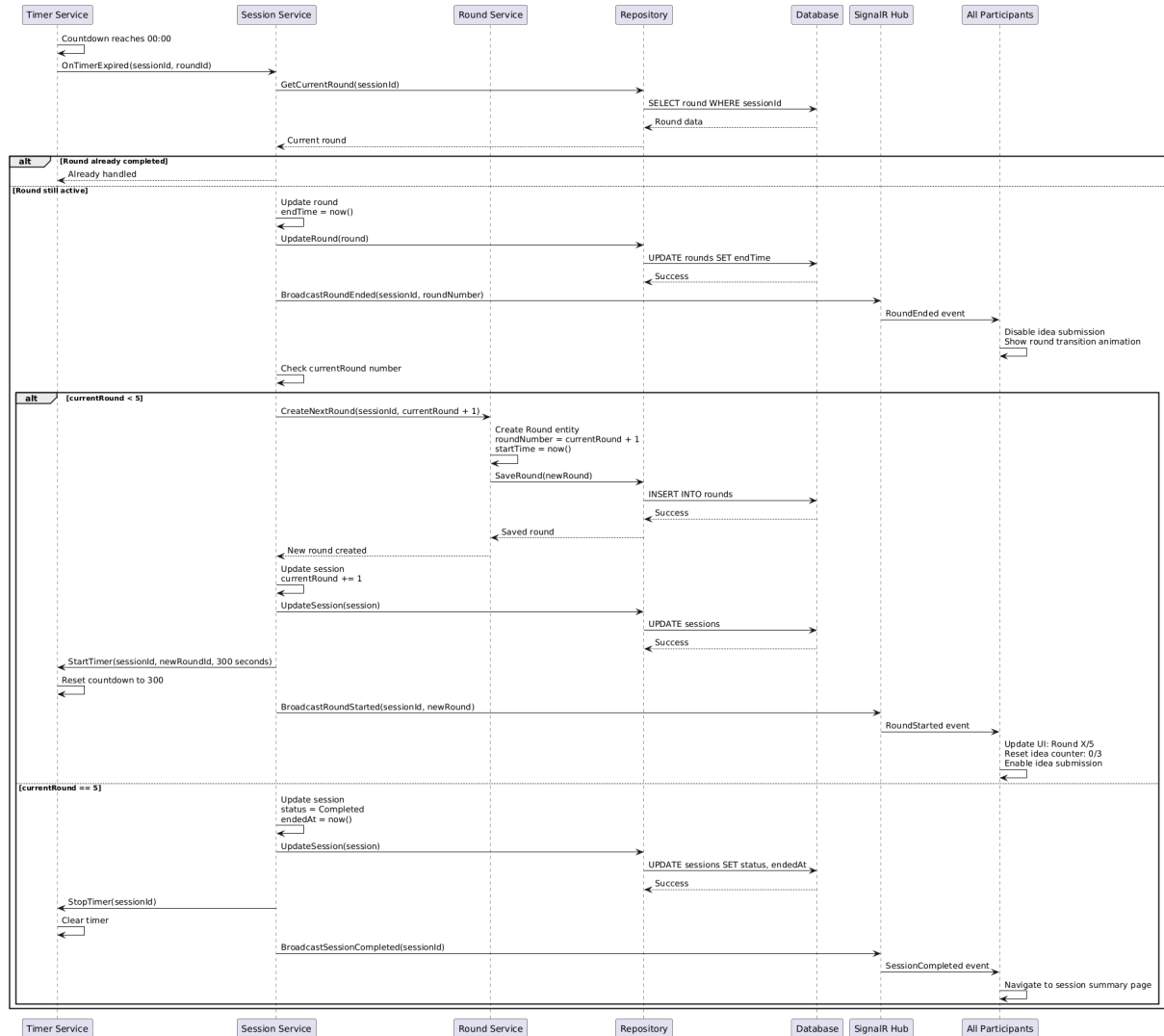
3.4.1.2 Start Brainstorming Session



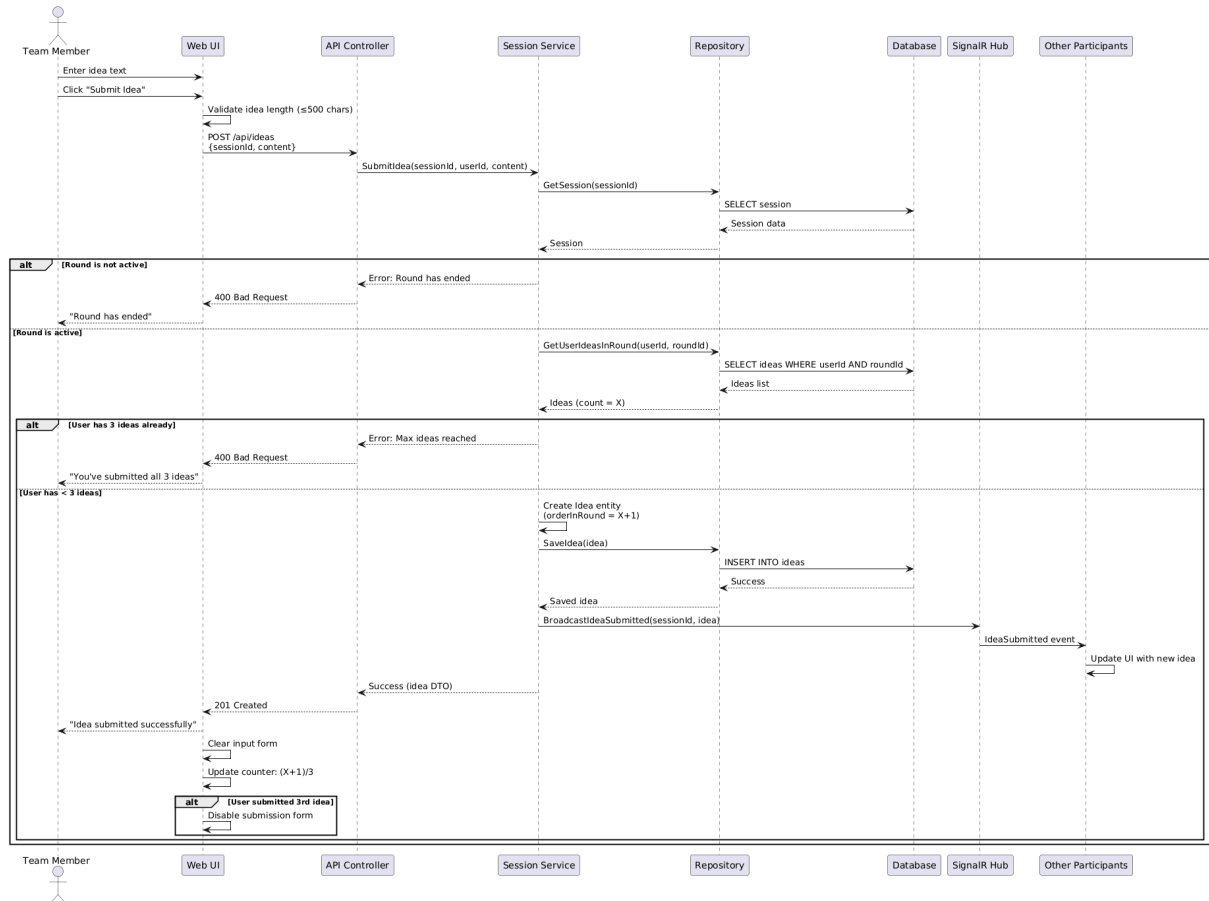
3.4.1.3 Create Team and Add Members



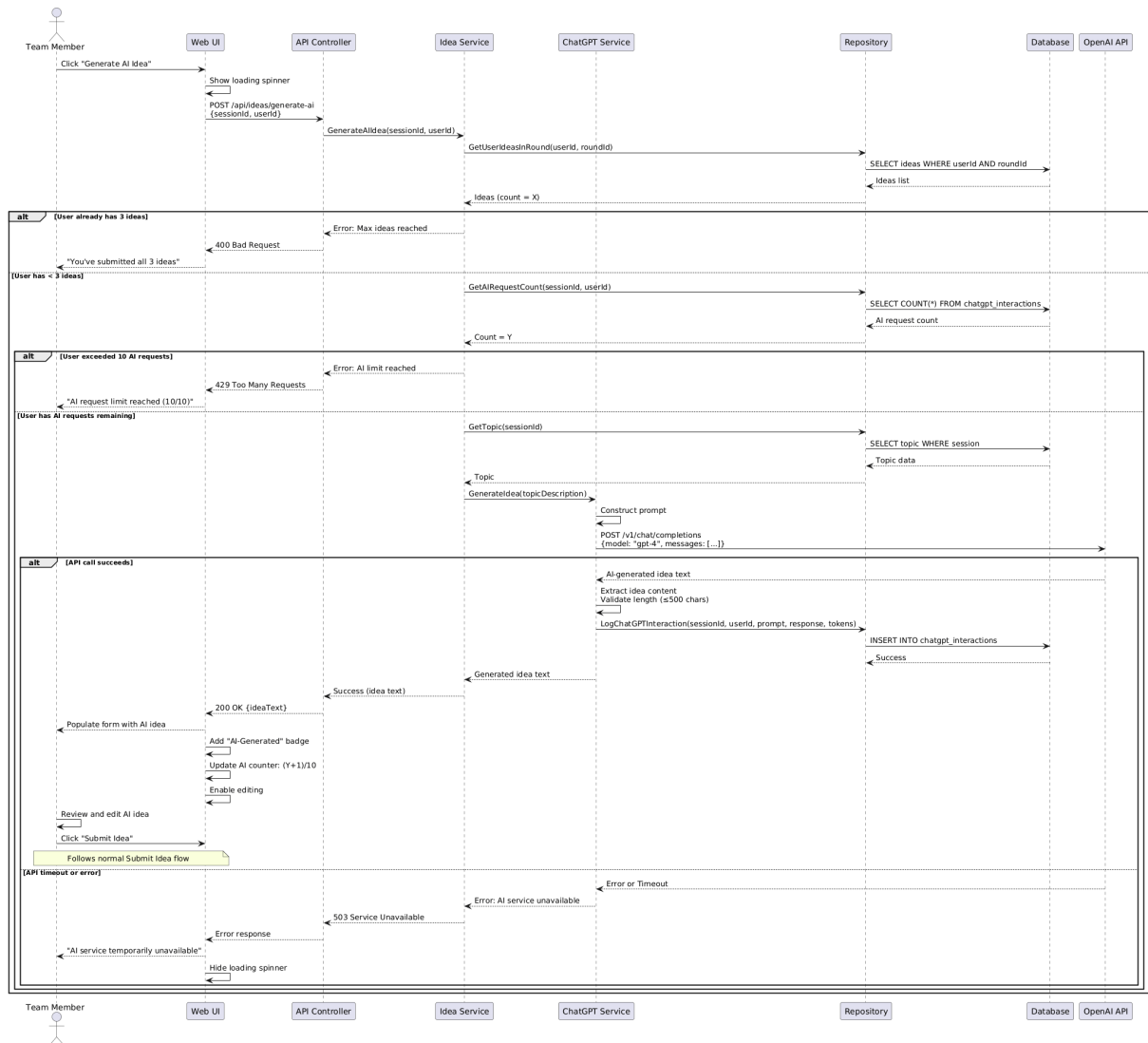
3.4.1.4 Automatic Round Advance (Timer Expiration)



3.4.1.5 Submit Idea During Active Session

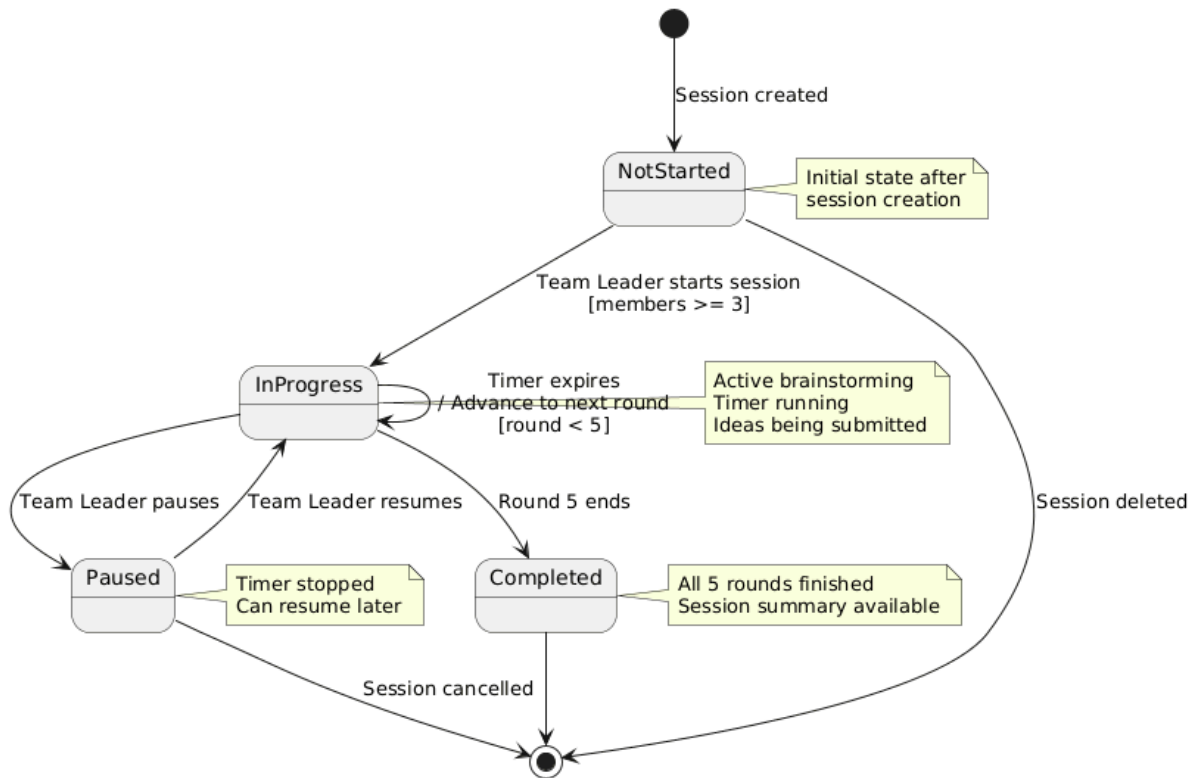


3.4.1.6 Request AI Idea Generator

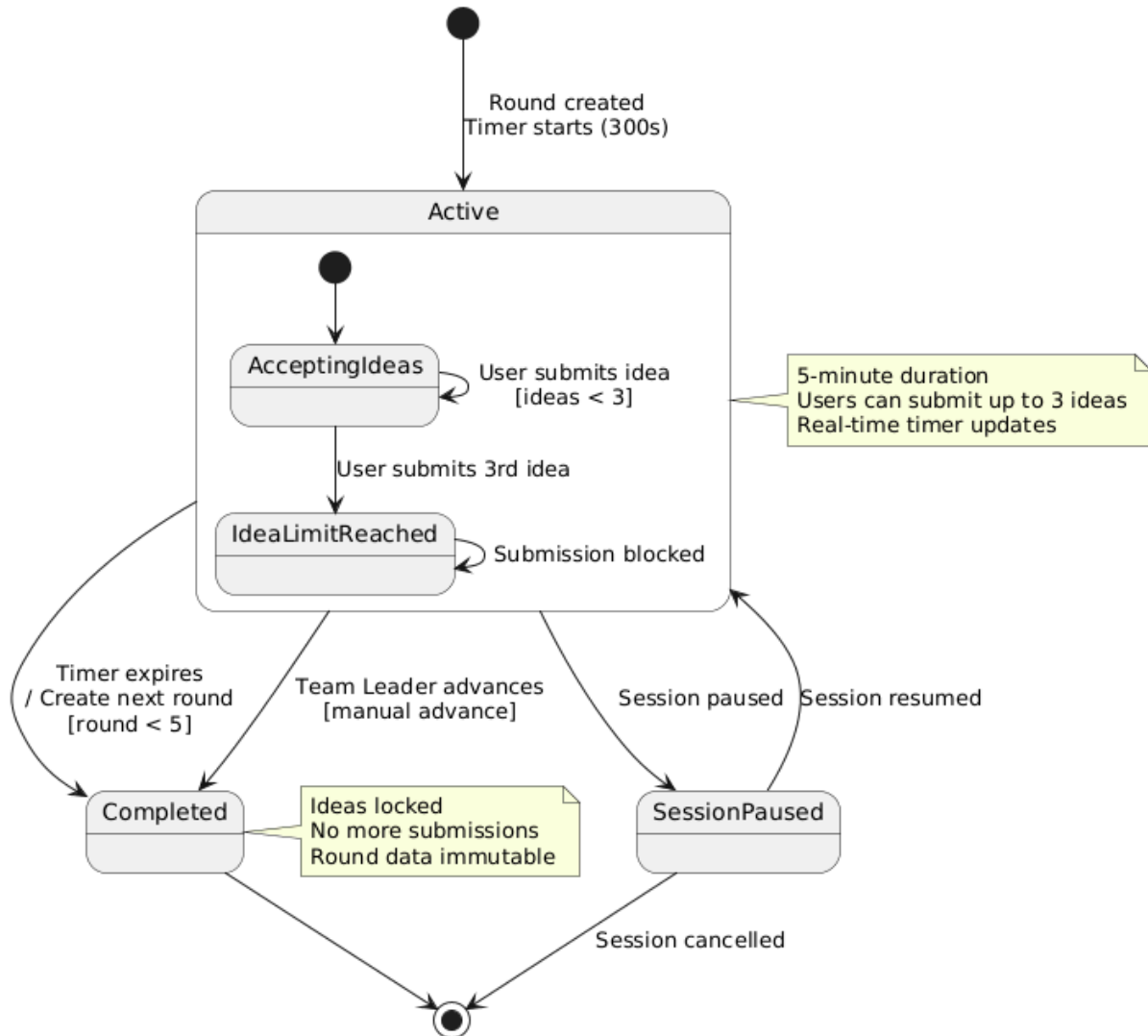


3.4.2 State Diagrams

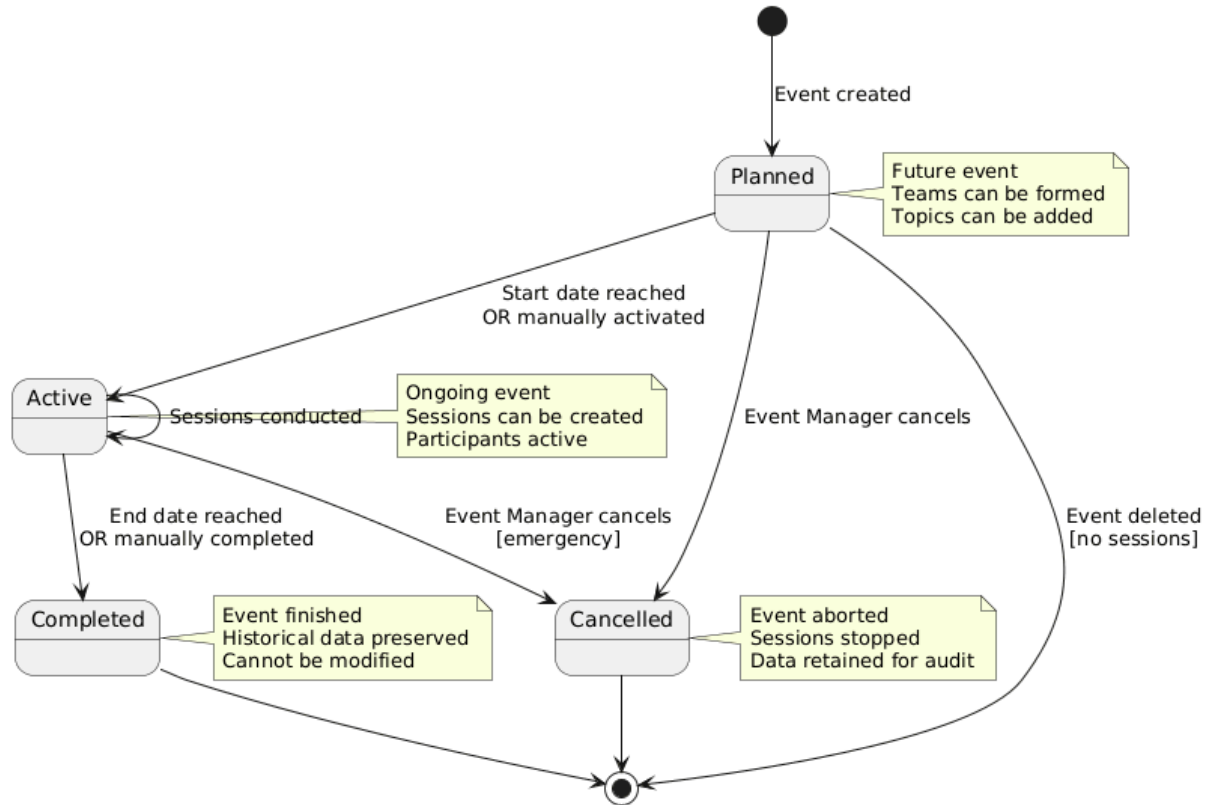
3.4.2.1 Brainstorming Session Lifecycle



3.4.2.2 Round Lifecycle



3.4.2.3 Event Lifecycle



■ **3.4.5 User interfaces and Mock-ups**

3.4.5.1 Authentication:


BrainStorm
Collaborative Brainstorming Platform

Welcome back
Sign in to continue your session

Continue with Google

Continue with Microsoft

OR

Email Address

Password

☐ Remember me

[Forgot password?](#)

Sign In

Don't have an account?

[Sign up](#)

3.4.5.2 Management Consoles:

Event Manager

12
Active

8
Planned

45
Completed

Create New Event

Set up your next event

Create Event

All Events

Tech Innovation Summit 2024
Jan 15 - Jan 17, 2024
8 participants
Manage Topics Assign Team

Annual Marketing Conference
Feb 20 - Feb 22, 2024
2 participants
Manage Topics Assign Team

Product Launch Event
Dec 10 - Dec 12, 2023
Completed

3.4.5.3 Team Console:

Team Console

Create New Team

Team Name

Team Description

Team Members (Max 6)

0/6

Member email

+ Add Member

Create Team

Current Team

Active

Innovation Squad
Product development and creative solutions team

Members (4/6)

Alex Chen
alex.chen@company.com

Sarah Wilson
sarah.w@company.com

Mike Rodriguez
mike.r@company.com

Add Member

3.4.5.4 Start New Session:

Start New Session

Launch a brainstorming session with your team

Session Topic

Select a topic

Session Duration

30 min 60 min 90 min

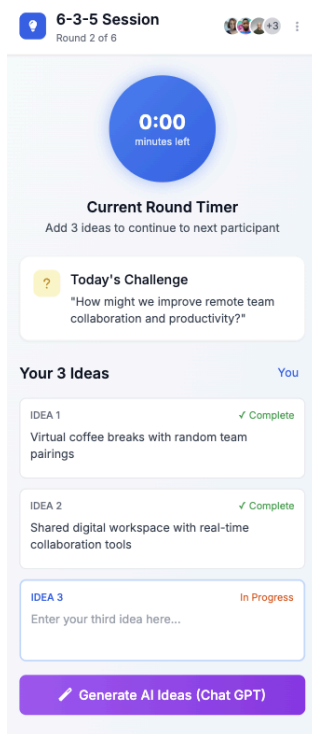
Start Brainstorming Session

Recent Sessions

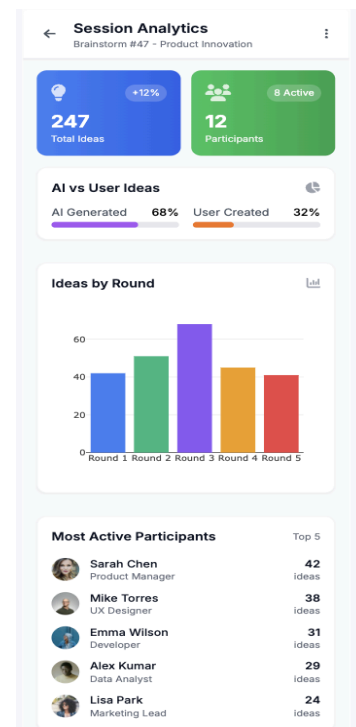
Product Features Brainstorm
2 hours ago • 45 min duration
Completed

Marketing Campaign Ideas
Yesterday • 60 min duration
Completed

3.4.5.5 6-3-5 Session:



3.4.5.6 Session Analytics



4 Glossary of Terms

- 4.1 **6-3-5 Method:** A structured brainstorming technique where 6 team members each write 3 ideas within a 5-minute cycle. The ideas are then passed to the next member in subsequent rounds to build upon.
- 4.2 **Actor:** A user or external system that interacts with the system. In this project, the primary actors are the Event Manager, Team Leader, and Team Member.
- 4.3 **CRUD:** An acronym for Create, Read, Update, and Delete, which represent the fundamental operations for data management.
- 4.4 **Cycle (or Round):** A single 5-minute time-boxed period within a 6-3-5 session during which a Team Member inputs ideas before passing them on.
- 4.5 **Event Manager:** A user role with administrative privileges, responsible for creating and managing events, topics, participants, and teams, as well as overseeing all sessions .
- 4.6 **FR (Functional Requirement):** A statement that specifies a function or behavior the system must be capable of performing .
- 4.7 **Ideathon:** A collaborative event, similar to a hackathon, where participants brainstorm and develop innovative solutions to specific problems or topics.
- 4.8 **NFR (Non-Functional Requirement):** A requirement that specifies criteria for the system's operation (e.g., performance, security, usability) rather than its specific behaviors.

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- 4.9 **RAD:** Requirements Analysis Document. The deliverable for this assignment, which defines what the system will do.
- 4.10 **Session:** A complete 6-3-5 brainstorming activity, initiated by a Team Leader, consisting of multiple rounds/cycles.
- 4.11 **Team Leader:** A user role responsible for managing a specific team, initiating and moderating 6-3-5 sessions, and interacting with AI features for the team .
- 4.12 **Team Member:** A user role that actively participates in brainstorming sessions by submitting and building upon ideas .
- 4.13 **UML (Unified Modeling Language):** A standardized modeling language used to visualize, specify, construct, and document the artifacts of a software system.
- 4.14 **Use Case:** A description of a specific interaction between an actor and the system to achieve a goal.

5 Traceability

Use Case (UC) List:

- **UC-1:** Manage Profile & Authentication
- **UC-2:** Manage Events
- **UC-3:** Manage Topics
- **UC-4:** Manage Teams
- **UC-5:** Manage Event Participation
- **UC-6:** Manage Brainstorming Session
- **UC-7:** Manage Rounds
- **UC-8:** Manage Ideas
- **UC-9:** Use AI Assistance
- **UC-10:** Receive Real-time Updates
- **UC-11:** Manage Session Logs
- **UC-12:** View Analytics & Reports

Traceability Matrix (Functional Requirements to Use Cases)

<i>Requirement Section</i>	<i>Requirement ID(s)</i>	<i>Requirement Description (Summary)</i>	<i>Mapped Use Case(s)</i>
User Management	<i>FR-1 to FR-10</i>	<i>User registration, login, profile management, and role-based access control [cite: 241-251].</i>	<i>UC-1</i>

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Event Management	<i>FR-11 to FR-20</i>	<i>Event CRUD operations (Create, Read, Update, Delete) and status management by Event Managers [cite: 252-262].</i>	<i>UC-2</i>
Topic Management	<i>FR-21 to FR-27</i>	<i>Topic CRUD operations, assignment to events, and status management by Event Managers [cite: 263-271].</i>	<i>UC-3</i>
Team Management	<i>FR-28 to FR-37</i>	<i>Team creation (max 6 members), member assignment, and team-to-event assignment [cite: 272-281].</i>	<i>UC-4</i>
Event Participation	<i>FR-38 to FR-43</i>	<i>Adding/removing participants to/from events; users viewing their events [cite: 282-289].</i>	<i>UC-5</i>
Session Mgmt	<i>FR-44 to FR-60</i>	<i>Team Leader creating, starting, pausing, resuming, and ending a 6-3-5 session [cite: 290-306].</i>	<i>UC-6</i>
Round Management	<i>FR-61 to FR-67</i>	<i>System automatically managing 5 rounds, each with a 5-minute timer and sequential numbering [cite: 307-314].</i>	<i>UC-7 (Often part of UC-6)</i>
Idea Management	<i>FR-68 to FR-81</i>	<i>Team Member submitting 3 ideas per round [cite: 316-318], editing</i>	<i>UC-8</i>

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		<i>ideas (pre-round end) [cite: 323], and viewing ideas [cite: 326-329].</i>	
ChatGPT Integration	<i>FR-82 to FR-91</i>	<i>Generating AI ideas [cite: 331], creating session summaries [cite: 337], and providing annotations [cite: 338].</i>	<i>UC-9</i>
Real-time Comms	<i>FR-92 to FR-102</i>	<i>Broadcasting session events (start, end, idea submit) via WebSockets [cite: 342-352].</i>	<i>UC-10 (Supports UC-6, UC-8)</i>
Logging & Audit	<i>FR-103 to FR-111</i>	<i>System logging all session actions (start, pause, idea submission) and providing logs to managers [cite: 354-362].</i>	<i>UC-11</i>
Reporting & Analytics	<i>FR-112 to FR-118</i>	<i>Displaying session analytics (total ideas, avg. per user) and exporting reports (PDF/Excel) [cite: 364-370].</i>	<i>UC-12</i>